INSTITUTO POLITÉCNICO NACIONAL
ESCUELA SUPERIOR DE CÓMPUTO

Selected Topics in Cryptography

Session 1: Elliptic curves *February 17, 2026*

Please do the following exercises by your own. Choose only one of the following programming languages to develop your code: C/C++, Python or Java.

1. Programming Exercises

1. Design and implement function that receives a prime number $p > 3$ as a paramaters a, b and find the values $a, b \in \mathbb{Z}_p$ for every non-singular elliptic curve $y^2 = x^3 + ax + b \mod p$. Use a library of the programming language to generate a prime number.

2. Design and implement a function that receives the size in bits of a prime number p and return the values $a, b \in \mathbb{Z}_p$ of a non-singular elliptic curve.

3. Design and implement a function that receives parameters $a, b, p > 3$ that define a non-singular elliptic curve E and compute and count the rational-points of the curve. Represent any point with three coordinates (x, y, z), $z = 1$ for every point different from infinity point, and $(0, 1, 0)$ for the point to infinity.

4. Design and implement a function that receives two different points P and Q in an elliptic curve E and return $P + Q$. Please use three coordinates to represent each point. Please do not join this function with the function to compute $2P$

5. Design and implement a function that receives a point $P \in E$ and returns $2P = P + P$. Please use three coordinates to represent each point. Please do not join this function with the function to compute $P + Q$, where $P \neq Q$.

2. Questions

Answer the following questions:

1. How many non-singular elliptic curves are for each prime number $p = 5, 7, 11$? Which of these curves have a prime number of rational points?
2. For three different non-singular elliptic curves, do the graphic using any online tool.
3. How many elliptic curves are singular for each prime number $p = 5, 7, 11$?
4. Generate a prime number for each size: 16 bits, 32 bits, 64 bits, 512 bits, 1024 bits and 2048 bits. For each prime number, find a non-singular elliptic curve.

3. Products

Every student must write a brief report, containing:

1. Your personal information, date of the lab session and the topic that we are studying in this lab session.
2. The source code you wrote to solve exercises 1-4, and a brief explanation of how it work.
3. Answers to questions of Section 2.

You must upload the pdf file of your report to Microsoft Teams.